Basic Statistics Cheat Sheet

1. Key Statistical Terms

- **Population**: The entire group being studied.
- **Sample**: A subset of the population used for analysis.
- **Variable**: A characteristic that can take different values (e.g., height, age, income).
- **Mean (Average)**: Sum of all values divided by the number of values.
- **Median**: The middle value when data is arranged in order.
- **Mode**: The most frequently occurring value.
- **Range**: The difference between the highest and lowest values.
- **Variance**: The measure of data dispersion around the mean.
- **Standard Deviation**: The square root of variance, indicating data spread.
- **Probability**: The likelihood of an event occurring.

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2. Types of Data

- **Quantitative Data**: Numerical data (e.g., height, weight, income).
- **Qualitative Data**: Categorical data (e.g., gender, color, brand names).
- **Discrete Data**: Countable numbers (e.g., number of students in a class).
- **Continuous Data**: Measurable values (e.g., weight, time, temperature).

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3. Types of Statistical Measures

- **Measures of Central Tendency**: Mean, Median, Mode.
- **Measures of Dispersion**: Range, Variance, Standard Deviation.
- **Correlation**: Measures the relationship between two variables.
- **Regression**: Predicts the relationship between dependent and independent variables.

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4. Types of Statistical Distributions

- **Normal Distribution**: Bell-shaped curve where most values cluster around the mean.
- **Binomial Distribution**: Used for binary outcomes (success/failure).
- **Poisson Distribution**: Used for counting the number of occurrences in a fixed interval.

- **Uniform Distribution**: All outcomes have equal probability.

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5. Hypothesis Testing

- **Null Hypothesis (H_0)**: Assumes no effect or no difference.
- **Alternative Hypothesis (H_1)**: Assumes a significant effect or difference.
- **p-Value**: Probability of obtaining the observed results if H_0 is true.
- **Confidence Interval**: Range in which the true population parameter lies with a certain probability.

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6. Common Statistical Tests

Test	Purpose
t-Test	Compares means of two groups
ANOVA	Compares means across multiple groups
Chi-Square Test	Tests independence between categorical variables
Pearson Correlation	Measures linear relationship between variables
Regression Analysis	Predicts dependent variable based on independent variables

7. Data Visualization Techniques

- **Histogram**: Shows frequency distribution.
- **Box Plot**: Displays median, quartiles, and outliers.
- **Scatter Plot**: Visualizes relationships between two variables.
- **Bar Chart**: Compares different categories.
- **Pie Chart**: Represents proportions.

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8. Statistical Software

- Excel
- R
- Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn)
- SPSS
- SAS